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Guohuan Su

suguohuan@ihb.ac.cn

Institute of Hydrobiology, Chinese Academy of Sciences

Kangshun Zhao

Institute of Hydrobiology, Chinese Academy of Sciences https://orcid.org/0000-0002-4302-3726

Julian Olden

University of Washington https://orcid.org/0000-0003-2143-1187

Jorge García Molinos

Hokkaido University https://orcid.org/0000-0001-7516-1835

Chunlong Liu

College of Fisheries, Ocean University of China

Min Zhang

College of Fisheries, Huazhong Agricultural University

Jun Xu

Institute of Hydrobiology, Chinese Academy of Sciences https://orcid.org/0000-0002-8065-296X

Otso Ovaskainen

University of Jyväskylä https://orcid.org/0000-0001-9750-4421

Biological Sciences - Article

Keywords:

Posted Date: January 19th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-8586370/v1

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Additional Declarations: There is **NO** Competing Interest.

Temperature regimes shape similarity of life-history strategies between native and non-native freshwater fishes

Kangshun Zhao[1,8], Julian D. Olden[2], Jorge García Molinos[3], Guohuan Su[4*], Chunlong Liu[5,6,7,8], Min Zhang[9], Jun Xu[1,10*], Otso Ovaskainen[11]

[1] State Key Laboratory of Lake and Watershed Science for Water Security, Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China.

[2] School of Aquatic and Fishery Sciences, University of Washington, Seattle, Washington, USA.

[3] Arctic Research Center, Hokkaido University, Sapporo, Japan.

[4] State Key Laboratory of Breeding Biotechnology and Sustainable Aquaculture, Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China.

[5] The Key Laboratory of Mariculture, Ministry of Education, College of Fisheries, Ocean University of China, Qingdao, China.

[6] Key Laboratory of Aquaculture Nutrition and Feed, Ministry of Agriculture and Rural Affairs, College of Fisheries, Ocean University of China, Qingdao, China.

[7] Shandong Key Laboratory of Green Mariculture and Smart Fisheries, College of Fisheries, Ocean University of China, Qingdao, China.

[8] Marine Environment and Resources Strategy Research Center, Ocean University of China, Qingdao, China

[9] Hubei Provincial Engineering Laboratory for Pond Aquaculture, Engineering Research Center of Green Development for Conventional Aquatic Biological Industry in the Yangtze River Economic Belt, College of Fisheries, Huazhong Agricultural University, Wuhan, China.

[10] State Key Laboratory of Marine Resource Utilization in South China Sea, Hainan University, Haikou, China.

[11] Department of Biological and Environmental Science, University of Jyväskylä, Jyväskylä, Finland.

[*] Corresponding author: Guohuan Su (suguohuan@ihb.ac.cn) and Jun Xu (xujun@ihb.ac.cn)

Abstract

Freshwater ecosystems are highly vulnerable to biological invasions, with increasing numbers of non-native fish species invoking severe ecological impacts. Life-history strategies reflect how fishes adapt to specific environments and play a critical role throughout the invasion process, leading to their application to understand large-scale patterns and drivers of freshwater fish invasions. Here, we compiled ecological traits for 14,679 freshwater fish species across 3,118 river basins worldwide to classify life-history strategies of native and non-native fishes and attempt to explain large-scale distributional patterns in diverse environments. We find that both native and non-native fishes are similarly shaped by environmental conditions (particularly temperature regimes) and exhibit consistent latitudinal gradients, with periodic strategists increasing and equilibrium and opportunistic strategists decreasing toward higher latitudes. Non-native fishes contain disproportionately more periodic and equilibrium strategists and fewer opportunistic strategists compared to native ones. Notably, the similarity of life-history strategies between native and non-native fishes is also primarily shaped by temperature regimes, increasing in colder or more seasonal environments but decreasing in warmer or more thermally stable ones. Overall, our results suggest that environmental filtering dominates in colder, more seasonal, high-latitude environments but weakens in warm, stable, low-latitude regions, where limiting similarity gains importance. This study provides new insights into how shifting environmental regimes shape global patterns in freshwater fish invasions and helps inform broad-scale conservation planning and management efforts.

Introduction

Biological invasions impose severe ecological disruptions on freshwater ecosystems and substantial socio-economic costs worldwide[1-3]. Freshwater ecosystems are highly susceptible to non-native fishes, experiencing biodiversity loss at rates exceeding those of terrestrial ecosystems[4-7]. The establishment of non-native fishes is strongly mediated by their intrinsic biological traits, which determine their capacity of pre-adaptation to environmental conditions in recipient ecosystems[8,9]. Among these, life-history traits play a critical role throughout the invasion process[1,10,11]. However, individual traits rarely act independently; instead, they covary through physiological and ecological trade-offs to form integrated life-history strategies that represent adaptations to specific biotic and abiotic environments[12,13]. Although freshwater fish invasions are intensifying worldwide, our understanding of the mechanisms and outcomes through the lens of life-history strategies remains limited.

The most widely applied life-history scheme for fishes is the equilibrium–periodic–opportunistic (EPO) model developed by Winemiller and Rose[14]. The EPO model classifies fish life-history strategies into three adaptive types based on trade-offs among juvenile survival, fecundity and generation time in response to environmental regimes[14,15]. Equilibrium strategists prevail in stable, predictable environments and are characterized by intermediate size, low fecundity, high juvenile survivorship, and late maturation. Periodic strategists are most prevalent in seasonal or periodically fluctuating environments and typically have large size, high fecundity, late maturation, and low juvenile survivorship. Finally, opportunistic strategists dominate in systems experiencing frequent and chaotic disturbances, having small size, relatively low fecundity, low juvenile survivorship, and short generation time (Supplementary Fig. S1). Compared to traditional trait-based classifications, functional groups, or the r/K-selection framework, the EPO model offers a more comprehensive representation of species' integrated ecological niches and differential adaptations to environmental conditions[12,13].

Life-history theory postulates that organisms adaptively vary the allocation of their energy resources among survival, growth, and reproduction to optimize fitness in diverse environments[12,16]. Therefore, environmental variability not only substantially affects the distributions of native and non-native fishes but also plays a crucial role in shaping their

morphology and life-history strategies[17-19]. Previous studies have shown that a combination of habitat conditions and climate has resulted in latitudinal gradients and biogeographical differences in fish morphological traits[20-22]. Although global analyses remain lacking, regional evidence suggests that similar factors could also shape the composition of native and non-native fish life-history strategies. For instance, rivers characterized by highly seasonal environments tend to be dominated by periodic strategists, whereas opportunistic strategists are favored in systems with greater flow variability[19,23,24]. In addition to environmental drivers, human activities may shape global patterns of life-history strategies in non-native fishes by nonrandomly introducing species with specific trait[25,26]. However, the large-scale geographical patterns of life-history strategies in native and non-native freshwater fishes and how environmental and anthropogenic factors shape them remain poorly understood.

Local or regional studies have sought to understand the mechanisms underlying invasion of non-native fishes through their life-history strategies, and have reported that invasive species often exhibit distinct life-history strategies compared to native species[19,27,28]. Although life-history theory provides a strong framework for understanding invasion success[29,30], the lack of global-scale analyses limits our ability to evaluate large-scale invasion mechanisms through the lens of life-history strategies. Two classical invasion hypotheses offer contrasting explanations for the mechanisms underlying the successful establishment of non-native species in novel environments—an issue known as Darwin's naturalization conundrum[31,32]. The 'limiting similarity hypothesis' (also known as 'Darwin's naturalization hypothesis') suggests that non-native species that are phylogenetically dissimilar to native species are more likely to naturalize due to reduced niche overlap and competition[33]. By contrast, the 'environmental filtering hypothesis' (or 'preadaptation hypothesis') proposes that non-native species that are more similar to native species are more likely to establish successfully because they are better pre-adapted to the local environmental conditions[34]. Yet neither hypothesis has consistently explained the distribution patterns of non-native fishes across global river basins. Moreover, Darwin evoked phylogenetic relatedness as an integrative surrogate for similarity in functional traits and niche preferences[31,32]; however, the lack of comprehensive trait data and methods for linking the traits to niche differences pose limitations[35], particularly for fish faunas. Therefore, clarifying global patterns and drivers of life-history strategy differences between native and

non-native fishes can provide essential insight into resolving Darwin's naturalization conundrum.

Here, we compiled a comprehensive global dataset of occurrence records and eight key life-history traits for 14,679 native and non-native freshwater fish species across 3,118 river basins using public databases and literature (Fig. 1a,b). This dataset enabled us to examine the global biogeographical patterns and the environmental and anthropogenic drivers of life-history strategy compositions for native and non-native fishes at the basin scale. We then evaluated the global-scale mechanisms of invasion in freshwater fishes by testing Darwin's naturalization conundrum within a life-history strategy framework. First, we expected that the life-history strategy compositions of native and non-native freshwater fishes would be shaped similarly by environmental conditions, potentially showing consistent biogeographical patterns (Fig. 1c). Previous studies highlight the role of environmental filtering in driving fish trait convergence patterns along climatic gradients and suggest that non-native fishes are more likely to occur where they are phylogenetically similar to native species[36-38]. Second, we expected that native and non-native fishes would differ markedly in their life-history strategy compositions at both global and biogeographical scales[19,27,28]. We further hypothesized that environmental filtering would dominate under more extreme environmental conditions, resulting in greater similarity of life-history strategies between native and non-native fishes, whereas more moderate or favorable environments would favor greater dissimilarity (Fig. 1d). Growing local and regional evidence suggests that these hypotheses may not be mutually exclusive and that their relative importance depends on environmental context[18,39-41]. Clarifying these patterns and their drivers is essential for advancing invasion theory and improving the prevention and management of freshwater fish invasions worldwide.

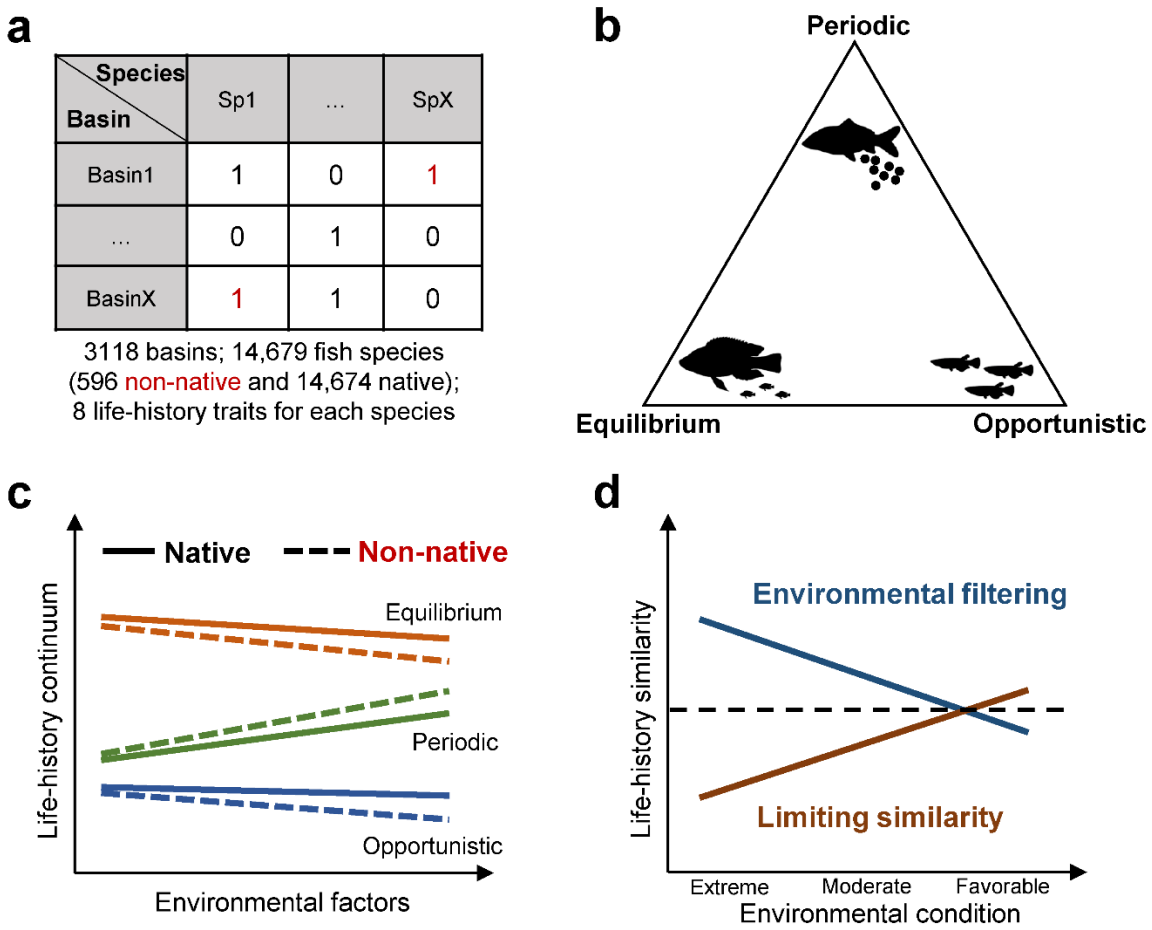


Fig 1. Conceptual diagram of the global dataset, life-history strategies, environmental responses, and invasion mechanisms in freshwater fishes. (**a**) Schematic representation of the global dataset (see Methods). (**b**) The three life-history strategies classified based on the eight life-history traits, following the classic EPO model (see Supplementary Fig. S1). (**c**) Life-history strategy compositions of native and non-native fishes are expected to show similar responses to environmental conditions, and potentially showing consistent biogeographical patterns. (**d**) Conceptual model illustrating how the relative importance of environmental filtering is expected to vary along environmental gradients by linking environmental conditions to the similarity in life-history strategies between two groups (see Methods). The central black dashed line marks where the strengths of the two mechanisms are equal.

Results

Latitudinal patterns of life-history strategies

The life-history traits of native and non-native fishes exhibited similar global latitudinal patterns in median values at the river-basin scale (Supplementary Figs. S2 and S3). With increasing latitude, both native and non-native fish species tended to show longer life span, delayed age at maturation, higher fecundity, larger body size, and declined growth rates. Consequently, both native and non-native fishes displayed consistent latitudinal gradients in the proportions of the three life-history strategies at the basin scale (Fig. 2a): the proportion of periodic strategists generally increased with latitude, while those of opportunistic and equilibrium strategists decreased. However, the rates of change along latitude differed between native and non-native species (Fig. 2b). In native fishes, the proportion of opportunistic strategists decreased more rapidly than that of equilibrium strategists (Supplementary Table S1), whereas the trend was reversed in non-native fishes. Moreover, the rate of increase in the proportion of periodic strategists with latitude was steeper in non-native fishes than in natives.

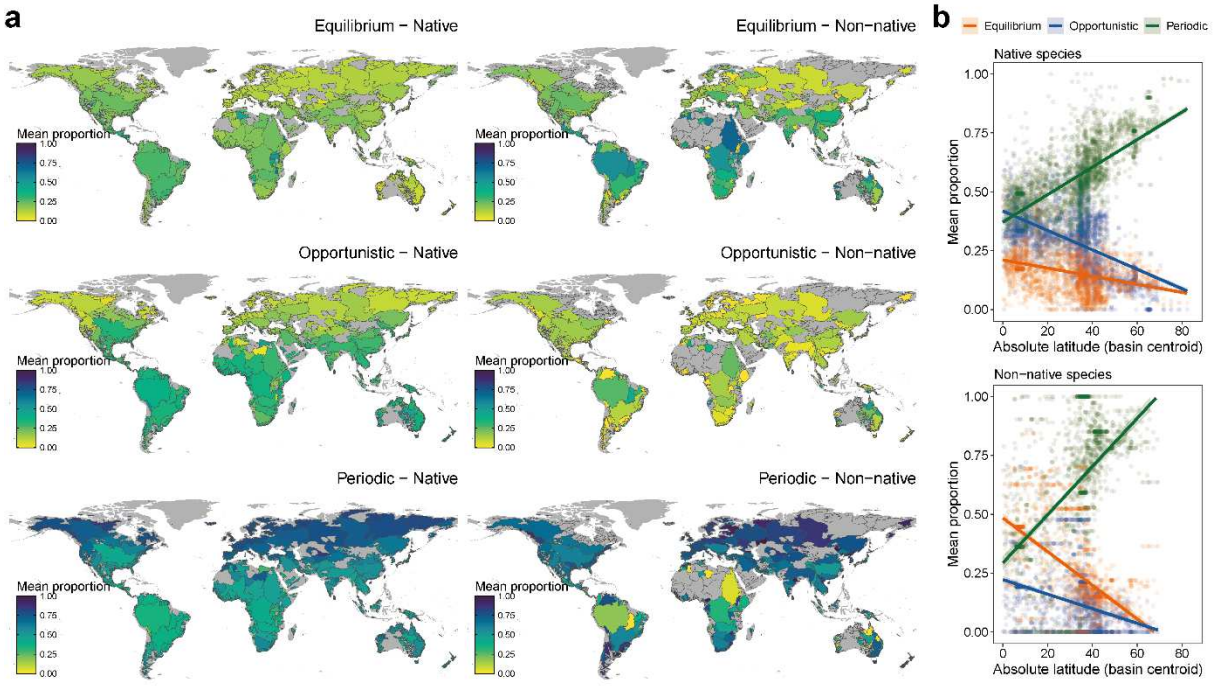


Fig 2. Global distribution patterns of life-history strategies of native and non-native freshwater fishes. (**a**) Geographic patterns of the mean proportions of the three life-history strategies of native and non-native fishes at the river-basin scale (see Methods). (**b**) Relationships between basin absolute latitude and the mean proportions of life-history strategies

for native and non-native fishes (for statistical comparisons, see Supplementary Table S1). The straight lines represent fits of a linear regression and error bands from the corresponding 95% confidence intervals.

Drivers of latitudinal patterns

Associations between life-history strategies and environmental factors were similar for both native and non-native fish species (Fig. 3). Specifically, with increasing mean annual temperature (MAT), the proportion of opportunistic and equilibrium strategists increased in both native and non-native fishes, whereas periodic strategists decreased. Temperature seasonality (TS) had similar effects on both groups, promoting periodic strategists and suppressing opportunistic and equilibrium strategists. In contrast, natural discharge (DIS) and the relative amplitude of discharge (RAD) had weaker associations compared to temperature, and differed slightly between native and non-native fishes (Supplementary Fig. S4). In native fishes, higher DIS was associated with an increase in equilibrium strategists and decreases in opportunistic and periodic strategists. In contrast, higher DIS was positively associated with opportunistic strategists while negative associated with equilibrium and periodic strategists for non-native fishes. RAD had similar effects on natives and non-native fishes, leading to higher proportions of opportunistic and equilibrium strategists and lower proportions of periodic strategists. In addition, increasing connectivity status index of river reaches (CSI) promoted native and non-native equilibrium strategists but suppressed opportunistic ones.

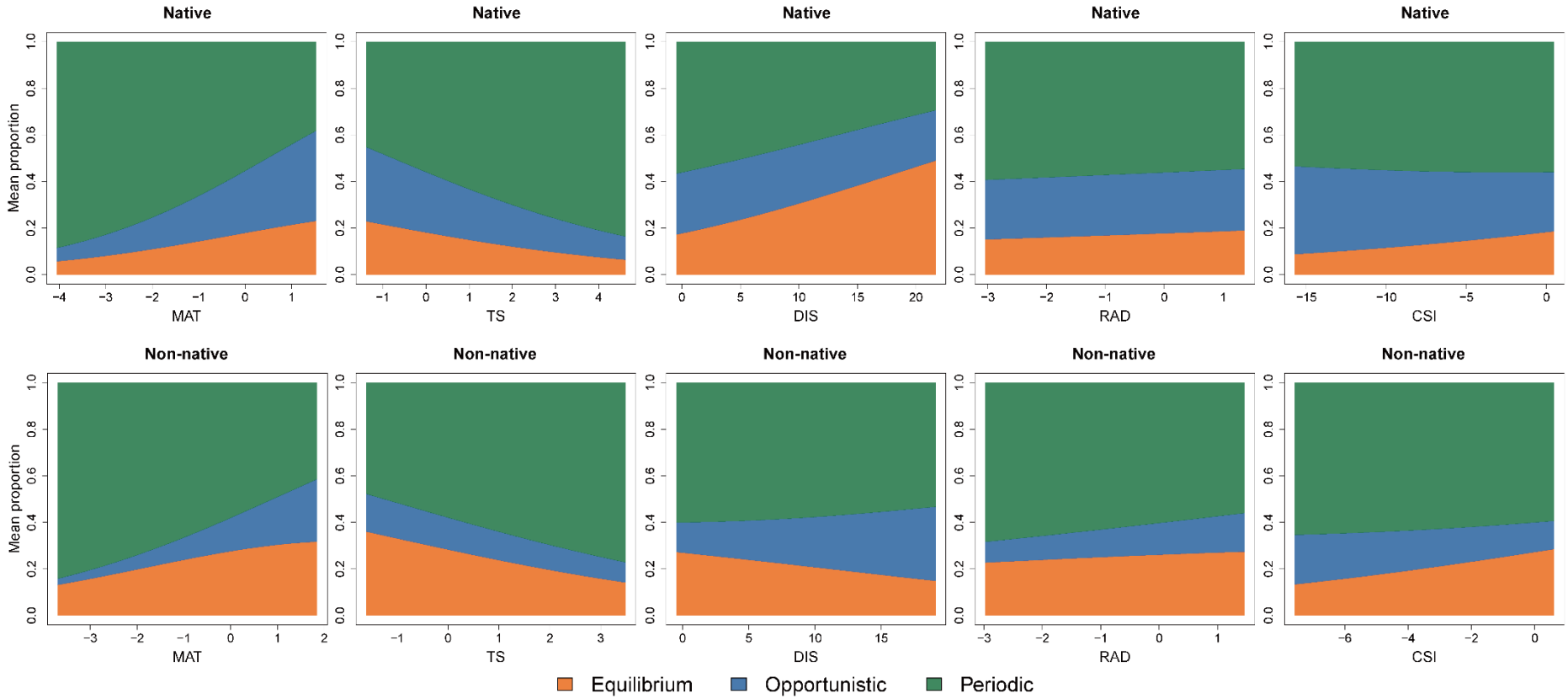


Fig. 3. Drivers of three life-history strategies in global native and non-native freshwater fishes. Each panel shows the model–predicted mean proportions of equilibrium, opportunistic, and periodic strategists along gradients of five key environmental and anthropogenic predictors at the basin scale: mean annual temperature (MAT), temperature seasonality (TS), natural discharge (DIS), relative amplitude of natural discharge (RAD), and connectivity status index of river reaches (CSI). The x-axis shows the standardized values of the focal predictors (z-transformed), and the y-axis shows the mean proportion of each life-history strategy, as predicted by the HMSC approach (n = 3,032 for the native group; n = 1,691 for the non-native group with all predictors complete). Colored stacked areas represent the three strategies, with their total height equal to 1 at any point along the predictor gradient. Effects of additional environmental and anthropogenic covariates are shown in Supplementary Fig. S5.

Differences in life-history strategies between native and non-native fishes

The global composition of the three life-history strategies showed significant differences between native and non-native freshwater fishes at the basin scale (Fig. 4a). Overall, native fishes were dominated by the periodic strategist followed by opportunistic and equilibrium strategists. The periodic strategist was also dominant among non-native fishes, followed by the equilibrium strategist and lastly the opportunistic strategist (Supplementary Table S2). Non-native fishes showed statistically significantly higher proportions of periodic and equilibrium strategists, but significantly lower proportions of opportunistic strategists compared to native fishes (Supplementary Table S3). This global pattern was largely consistent across the six biogeographic realms, with only the Indo-Malay and Nearctic realms showing significantly lower proportions of periodic and equilibrium strategists, respectively, in non-native fishes compared with native ones (Fig. 4b).

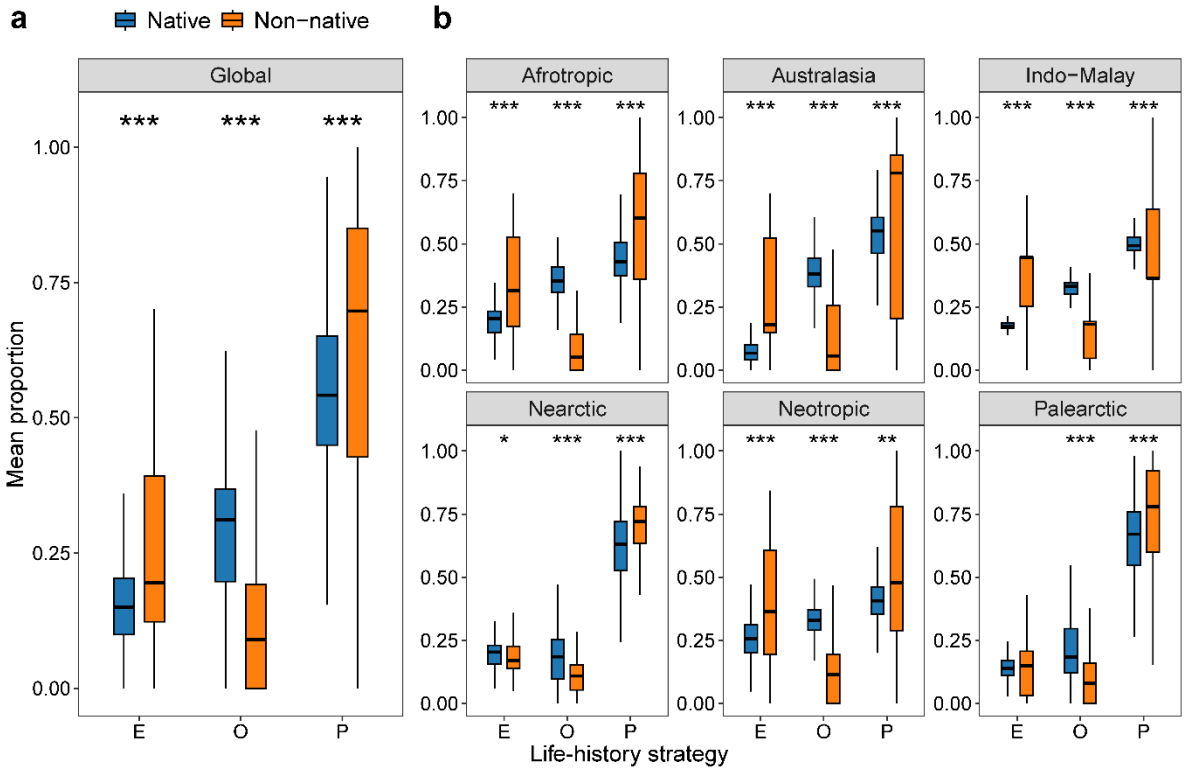


Fig. 4. Comparison of life-history strategy compositions between native and non-native freshwater fishes. Boxplots of the mean proportions of equilibrium (E), opportunistic (O), and periodic (P) strategies of native and non-native fishes at the basin scale for (**a**) global and (**b**) biogeographical realms. The stars above the boxplots indicate statistically significant differences between the two groups (Wilcoxon rank sum test *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; for the complete test results see Supplementary Table S3). The global distribution of the six biogeographical realms is shown in Supplementary Fig. S6.

Similarity patterns of life-history strategies and drivers

Similarity in life-history strategies at basin scale between native and non-native fishes exhibit a marked global latitudinal gradient (Fig. 5a,b). In the Northern Hemisphere, similarity increased with latitude, rising sharply from temperate regions (~30°N) and approaching one in Arctic basins. In the Southern Hemisphere, it decreased from tropical to temperate latitudes but increased again beyond ~30°S, although remaining lower than in the north. Overall, similarity tended to increase with latitude, indicating that native and non-native fishes shared more similar life-history strategies at higher than at lower latitudes. Both mean annual temperature and temperature seasonality significantly affected similarity although in opposite ways (Fig. 5c): similarity decreased with increasing temperature but increased with greater temperature seasonality. Meanwhile, the negative effect of mean annual temperature on similarity was slightly stronger in high-discharge basins (Supplementary Table S4). These results suggest that the relative importance of the two competing hypotheses in the context of Darwin's naturalization conundrum varies systematically along latitudinal environmental gradients (Fig. 5d). Higher similarity between native and non-native fishes suggests that environmental filtering plays a stronger role in colder and more seasonally variable high-latitude regions (Fig. 5a and Supplementary Fig. S7a,b). Conversely, the lower similarity observed in warmer and more stable low-latitude regions suggests a weakening of the effect of environmental filtering replaced by an increasing influence of the limiting similarity hypothesis. In addition, higher native species richness and GDP had significant positive effects on similarity (Supplementary Table S4).

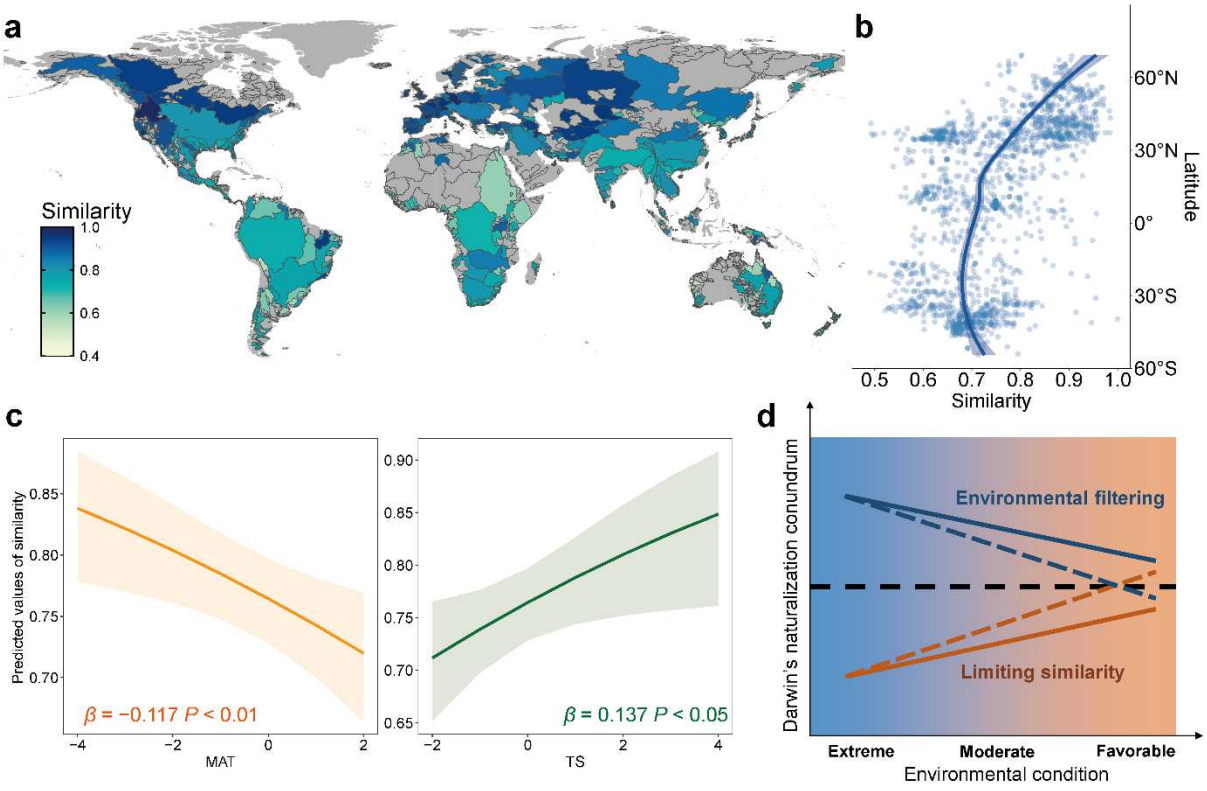


Fig. 5. Global patterns and drivers of life-history strategy similarity between native and non-native freshwater fishes. (**a**) Geographic distribution of similarity between native and non-native fishes across 1,720 river basins (see Methods). (**b**) Relationship between basin central latitude and native–non-native similarity. Curve represent fitted trends based on LOESS smoothing. (**c**) Marginal effects (model predictions) of mean annual temperature (MAT) and temperature seasonality (TS) on native–non-native similarity (Supplementary Table S4). Note that predictors are standardized by subtracting the mean and dividing by the standard deviation. (**d**) Conceptual diagram illustrating changes in the two hypotheses of Darwin's naturalization conundrum along environmental gradients. We treat cold, highly seasonal, high-latitude climates as extreme conditions and warm, less seasonal, low-latitude climates as moderate-to-favorable conditions, because global patterns of freshwater fish richness indicate that diversity peaks in low-latitude regions with high MAT and low TS (Supplementary Fig. S8a). Our results suggest two plausible scenarios: (i) environmental filtering consistently outweighs limiting similarity across the entire environmental condition, including in moderate or favorable low-latitude environments (solid lines); (ii) limiting similarity can surpass environmental filtering under moderate to favorable conditions, although environmental filtering remains dominant globally (dashed lines). The central black dashed line marks where the strengths of the two mechanisms are equal. All error bands represent 95% confidence intervals.

Discussion

This study provides the first global synthesis of life-history strategies in native and non-native freshwater fishes and reveals three major findings. First, both groups showed parallel latitudinal gradients in life-history strategies (periodic strategists increasing toward higher latitudes and equilibrium and opportunistic strategists decreasing) and responded similarly to climatic and hydrological drivers. Second, non-native fishes systematically diverged in life-history strategy composition, containing disproportionately more periodic and equilibrium strategists and markedly fewer opportunistic strategists than native fishes. Third, the similarity of life-history strategies between native and non-native fishes at the basin scale varied predictably along climatic gradients, increasing in colder and/or more seasonal environments (high latitudes) but decreasing in warmer and/or more thermally stable regions (low latitudes).

Both native and non-native freshwater fishes exhibit consistent latitudinal gradients in life-history strategies (Fig. 2). Specifically, the prevalence of periodic strategists increased toward higher latitudes, whereas equilibrium and opportunistic strategists decreased. These convergent gradients likely reflect the pervasive influence of climatic conditions (e.g., temperature regimes) in shaping life-history traits and strategies of both native and non-native fishes (Supplementary Figs. S2 and S3)[22,42,43]. For instance, high-latitude fish populations generally grow more slowly, mature later, attain longer longevity, and reach larger sizes than those at lower latitudes[44]. Fish species with low fecundity but high juvenile survival through parental care (typical equilibrium strategists) tend to be favored in more benign environmental regimes and higher productivity, such as warm climates[38,45]. However, the proportions of the three life-history strategies differed between native and non-native fishes at the same latitude, while the rates of change in strategy composition with latitude also diverged (Fig. 2b). This may reflect biases toward life-history strategies of non-native fishes that confer advantages under different abiotic or biotic conditions (see below for more details).

Across latitudinal gradients, variation in freshwater fish life-history strategies was closely related to temperature regimes and hydrological conditions. Both native and non-native fishes showed consistent responses to temperature: periodic strategists dominated in cold environments, whereas opportunistic and equilibrium strategists were more common in warmer

regions (Fig. 3). Cold environments typically favor species with larger body size, delayed maturation, and longer life span[42,43], whereas high temperatures reduce resource limitation and juvenile density dependence, promoting opportunistic and equilibrium strategists[38,46]. Moreover, greater temperature seasonality are hypothesized to favor periodic strategists over opportunistic and equilibrium strategists, supporting the prediction that opportunistic species are favored in environments characterized by frequent and intense disturbances, whereas seasonal environments are more suitable for periodic species[14,47]. Mean annual discharge also played a key role, but showed contrasting effects between native and non-native species (Fig. 3). In native fishes, higher discharge generally indicated greater habitat stability and resource availability, favoring equilibrium strategists adapted to predictable environments. In contrast, for non-native species, high discharge likely served as a hydrological disturbance that promoted opportunistic species capable of rapid colonization and reproduction[23,48]. Increasing amplitude of discharge enhanced the prevalence of opportunistic strategists and suppressed periodic ones in fishes, consistent with life-history theory and previous regional studies[23,24,49]. Finally, higher river connectivity promoted native and non-native equilibrium strategists, implying that less habitat fragmentation can lead to more stable environments[50]. In general, our results suggest that temperature exerted a stronger control on global life-history patterns, while hydrological conditions and connectivity further modulated these responses through their influence on habitat predictability and disturbance regimes (Supplementary Fig. S4).

Despite their shared latitudinal gradients, native and non-native fishes differed markedly in their strategy compositions (Figs. 2 and 4a). Globally, non-native species were disproportionately dominated by periodic and equilibrium strategists while opportunistic strategists were consistently underrepresented compared with native species. This general pattern was largely consistent across the six biogeographic realms (Fig. 4b). Opportunistic strategists, which are typically small-bodied, fast-growing, and early-maturing, are often associated with rapid colonization ability[14,47]. By contrast, periodic and equilibrium species generally require longer times to establish self-sustaining populations. Therefore, the global dominance of periodic and equilibrium strategists among non-natives likely reflects human-mediated introduction pathways that preferentially select for species with particular life-history traits[51,52]. Most non-native fishes have been introduced through aquaculture, trade, and

recreational fisheries[1,53]. Species with large body size, high fecundity, generalist diets, and high trophic levels (i.e., periodic strategists) are particularly favored because of their high economic value[54-56]. For instance, the common carp (*Cyprinus carpio*), a classic periodic strategist, is one of the most widely established non-native species globally. Similarly, equilibrium strategists with high parental care are often introduced and established, as their enhanced offspring survival increases establishment success and their reproductive traits make them easier to breed in captivity[26,28,55]. Meanwhile, aquaculture escapes frequently occur in high numbers, further facilitating establishment of periodic and equilibrium strategists[57].

Even under targeted human-mediated introductions, we found that native and non-native fishes exhibited consistent latitudinal gradients in life-history strategies and responded similarly to environmental drivers (Figs. 2 and 3). This suggests that the establishment of non-native species is predominantly governed by environmental filtering. Previous studies have reported that non-native fishes are more likely to establish when they are phylogenetically like native species[36,37]. However, the observed differences in the rates of change in strategy proportions along latitude may indicate that environmental filtering does not operate uniformly across climatic gradients (Fig. 2b). Our analysis of similarity patterns confirmed this context dependence (Fig. 5a). Non-native species in colder and more seasonal high-latitude regions exhibited life-history strategies more similar to native species suggesting prevalence of environmental filtering processes. Conversely, species in warmer and less seasonal low-latitude regions were more divergent pointing towards dominance of competitive exclusion through competition (i.e., limiting similarity) (Fig. 5b,c and Supplementary Table S4). This pattern aligns with the 'Latitudinal Biotic Interaction Hypothesis' that biotic interactions weaken from low to high latitudes[58], where thermal constraints on growth and reproduction also intensify[38]. Consequently, the relative importance of these two mechanisms shifts predictably with climatic context (Fig. 5d), providing a plausible explanation to Darwin's naturalization conundrum similar to findings from plant communities[31,41,59]. Moreover, higher native species richness and GDP increase similarity, likely because species-rich basins already occupy most of the life-history strategy space, and only non-native species with similar strategies can establish. In high-GDP regions, intensive human-mediated introductions further increase the likelihood that introduced species resemble native species. Collectively, our results highlight that

environmental filtering remains the dominant process facilitating the establishment of non-native freshwater fishes globally, but its influence weakens in low-latitude, warm and stable environments where limiting similarity becomes increasingly important.

These insights carry several implications for the management of global freshwater fish invasions. First, high-latitude regions may face greater invasion risks from periodic strategists, while low-latitude systems may be more vulnerable to invasions by equilibrium and opportunistic fish species. Second, although opportunistic strategists were least represented among non-native fishes, their strong colonization and dispersal potential warrant further consideration as potential future invasives[60], particularly in low- and mid-latitude regions. These findings underscore the importance of developing region-specific management strategies tailored to the predominant invasion risks associated with different life-history strategies[1]. Third, given projected global change, shifts in temperature regimes are likely to alter the latitudinal structure of invasion strategies and the balance between environmental filtering and limiting similarity effects, with uncertain consequences for invasion[61,62]. For example, climate warming may expand the potential ranges of equilibrium and opportunistic invasive fishes while weakening the establishment potential of periodic invasive fishes at higher latitudes. Furthermore, climate change is also reducing hydrological seasonality of high-latitude rivers, potentially weakening the advantages of periodic strategists[63]. Such directional and regionally heterogeneous changes, which are not explicitly captured in our models, may already be influencing the global distribution of life-history strategies and native–non-native similarity and are expected to modify invasion risk through time further. Therefore, despite the complexities of managing native fishes with disparate life histories and varying susceptibilities to non-native species, integrating life-history theory into conservation and invasion management provides an alternative tool for prioritizing management efforts across climatic zones.

Methods

Distribution data of global freshwater fish species

We employed a highly comprehensive global database of freshwater fish species distributions compiled by Tedesco et al.[64]. This database provides complete fish species lists for 3,119 drainage basins covering more than 80% of the Earth's surface, including the occurrence of 14,953 fish species inhabiting permanently or occasionally freshwater systems. Documented fish occurrences in each river basin account for all freshwater species inhabiting the entire river network of each basin, from first order streams to the sea. Each species in each river basin is assigned a native or non-native status, with a non-native species defined as one that has been introduced, completed its life cycle and established a self-sustaining population in that basin. The non-native species roughly corresponds to the postindustrial period (i.e., after the 18th century) when industrialization began and fish introductions for aquaculture, fishing, and ornamental purposes sharply increased[5,43,64,65]. Each river basin in the database is nested within seven biogeographical realms (i.e., Afrotropical, Australasia, Indo-Malay, Nearctic, Neotropic, Oceania, and Palearctic)(Supplementary Fig. S6)[66]. Geographic coordinates of the centroid and the surface area of each basin are also provided, which we used to locate all river basins across the seven biogeographic realms. Due to the small number of basins in Oceania, it was merged with Australasia for analysis. We verified the validity of all fish species names according to FishBase using R package '*rfishbase*'[67]. The resulting occurrence matrix included 14,862 valid species names, after removing 91 duplicate or invalid entries from the initial species list.

Trait data

Eight life-history traits were selected to describe the theoretical life-history strategies of fish species[13,48,68]. These eight traits were life span (maximum potential age in years), maximum total length (cm), parental care (categorized as non-guarders, guarders, and bearers), age at maturation (mean age at sexual maturity in years), fecundity (average total number of eggs or offspring per reproductive cycle), offspring size (length at birth in millimeters), trophic level (trophic position within the food web), and the von Bertalanffy growth coefficient (K, representing the annual individual growth rate).

We used public databases and scientific literature to compile a comprehensive life-history trait dataset for global freshwater fishes. First, we obtained data on life span, maximum total length, parental care, age at maturation, fecundity, trophic level, and the K from FishBase[67,69]. Age at maturation prioritized female mean values but use male values where female data were not available. Trophic level and K were fully available across species and parental care was transformed from categorical to continuous values, using a similar approach to Winemiller[47], as follows: (1) non-guarders, (2) guarders, and (3) bearers. Then, we integrated the Fishbase data for four traits (i.e., life span, maximum total length, age at maturation, fecundity) with the life-history trait datasets for global freshwater fishes (6,293 species) and North American freshwater fishes (809 species) to maximize the proportion of available trait information[52,70]. Specifically, we retained the maximum value for life span and maximum total length across the three sources while we took the mean value for fecundity and age at maturation. Finally, we supplemented our trait dataset using the R package '*FishLife*'[71], from which all data for offspring size were obtained. Overall, 17.5% of trait values were missing across all species in the resulting trait dataset, with missing rates per individual trait ranging from 0% for maximum total length to 25.6% for offspring size (Supplementary Table S5), which were subsequently estimated using the imputation approach described below.

Trait data imputation

Phylogenetic imputation is a valuable approach for estimating missing life-history traits[72,73]. We statistically imputed missing trait values with a phylogenetically informed predictive model using R package '*Rphylopars*'[73]. We constructed a global phylogenetic tree for the 14,862 valid freshwater fish species using the R package '*FishPhyloMaker*'[74,75]. A total of 183 species names were excluded as unidentifiable during the process. Consequently, we obtained a clean occurrence dataset with phylogenetic information for a final set of 14,679 species (596 non-native and 14,674 native) across 3,118 river basins. We applied a natural logarithmic transformation to all continuous valued traits to better fit the assumed distribution and to avoid obtaining negative values after the data imputation. We then determined the best-fitting phylogenetic informed predictive model based on the Akaike Information Criterion for a set of candidate models including Brownian motion, Pagel's, Ornstein-Uhlenbeck, and Early-Burst

models (Supplementary Table S6). We subsequently tested the accuracy of the best performing model (Brownian motion) in filling in missing values on a set of species with original complete information for all traits (9,691 species out of 14,679). We randomly removed 20% of their trait values, excluding maximum total length and trophic level which information was available for all species (Supplementary Table S5), and imputed them using the selected best phylogenetic model. Imputation accuracy was subsequently calculated using the Spearman correlation between the imputed and observed values. This procedure was repeated 100 times to ensure robust estimation of the Spearman correlation coefficient, which ranged from 0.88 to 0.96 (Supplementary Table S7). As a comparison, we alternatively imputed the missing values with the classical method of using the average trait value of closely related species (by genera), but the phylogenetically informed predictive model outperformed this alternative method (Supplementary Table S7). Therefore, we imputed missing values using the Brownian motion model.

Classification to life-history strategies

Following Winemiller and Rose[14], we classified the 14,679 fish species into three primary strategies: equilibrium, periodic, and opportunistic. We then used the unsupervised learning method of archetype analysis (AA) to characterize life-history strategies for each fish species, following the approach of Pecuchet et al.[13]. AA focuses on identifying extreme values that can be used as archetypes rather than on the means or medians of a cluster. Therefore, the method can quantify species' positions within the trilateral continuum by estimating their relative distances to opportunistic, periodic, and equilibrium-based endpoints (i.e., archetypes), yielding proportional life-history strategy scores[76]. AA offers advantages over traditional clustering methods such as k-means and k-medoids, which may arbitrarily assign neighboring points to distinct clusters when trait data are continuous. It is well-suited for detecting atypical species and characterizing life-history strategies based on traits[13,77,78].

We performed soft clustering using the "*stepArchetypes*" function from the R package '*archetypes*', with the argument family = robust() to reduce the influence of the outliers[78]. Archetype analysis was first applied to the trait matrix with a fixed number of strategies (k = 1 to 10) and the residual sum of squares (RSS) was computed over 10 repetitions for each k[79].

Visual inspection of the residual sum of squares (RSS) suggested that the 'elbow criterion' did not clearly indicate an optimum; we selected k = 3 to align with the EPO model (Supplementary Fig. S9). The results of the AA, particularly the positions of the archetypes in trait space, were visualized using a principal component analysis (PCA) biplot based on the eight life-history traits (Supplementary Fig. S10). For each species, the proportions associated with the three life-history strategies sum to one. The proportion of each life-history strategy per basin was calculated as the sum of species-specific weights for that strategy divided by the total weights of all species in the basin (i.e., mean value per life-history strategy), computed separately for native and non-native species.

Each extreme point in the trait space corresponds to a unique and functionally distinct life-history strategy (Supplementary Fig. S10). These strategies and their relations with traits corresponded closely to the theoretical model of Winemiller and Rose[14] and are henceforth referred to as equilibrium, periodic, and opportunistic strategies, respectively (Supplementary Fig. S1). Equilibrium strategists are characterized by small to medium in body size, high parental care, low fecundity, large offspring size, and late maturation. They rely on stable and predictable environments with low environmental variation and strong biotic interactions. Periodic strategists display long life spans, large body size, late maturation, high fecundity, and low parental care (i.e., low juvenile survivorship). They are most prevalent when the environmental variability impacting early life stages is periodic and large scale. Opportunistic strategists are usually small in size, short-lived, fast-growing, and mature early, with relatively low fecundity and low parental care. They are adapted to highly variable, unpredictable, disturbed, and ephemeral habitats, and are characterized by rapid colonization ability[14,60].

Similarity in life-history strategies

To quantify the similarity in life-history strategies between native and non-native fishes within each basin, we projected species into a ternary life-history strategy space defined by the three endpoints: equilibrium (E), opportunistic (O), and periodic (P) (Supplementary Fig. S11). The position of each species in the ternary plot was estimated based on eight life-history traits following the EPO model of Winemiller and Rose[14]. For each basin, we calculated the centroids of native and non-native fishes, defined as the arithmetic mean of species' coordinates in the

ternary space. The centroid distance (*cd*) was computed as the Euclidean distance between the two centroids, representing the separation between the non-native species pool and the recipient (native) community[80]. A larger *cd* value indicates a lower degree of overlap in life-history strategies between the non-native and native fishes within the basin, whereas a smaller value suggests greater overlap. To provide a standardized measure of similarity (*sim*), we further rescaled the centroid distance into the range (0, 1], with values close to 1 indicating high similarity and values approaching 0 indicating low similarity (Supplementary Fig. S11). *sim* was computed for all 1,720 basins containing at least one non-native and one native species.

Environmental and anthropogenic variables

We selected environmental and anthropogenic variables that are widely associated with fish life-history traits and strategies[19,23,49,81,82]. The environmental variables included mean annual temperature (MAT), temperature seasonality (TS), natural discharge (DIS), relative amplitude of natural discharge (RAD), native species richness (NSR), and river basin area (RBA). The anthropogenic variables included the connectivity status index of river reaches (CSI) and gross domestic product (GDP). Among them, NSR, RBA, and GDP were included as covariates to account for potential effects of differences in species richness, habitat size, and variation in introduction pressure on the composition and similarity of life-history strategies of native and non-native fishes. Specifically, MAT and TS (both at a resolution of ~1 km², 30 arc-seconds) were obtained from the WorldClim 2.0 database (www.worldclim.org)[83]. Air temperatures (i.e., MAT and TS) were used as a proxy for water temperatures, as it has been demonstrated that air and river water temperatures are strongly correlated particularly at the temporal resolution (annual means) considered here[84]. Annual average DIS was obtained from the HydroSHEDS database (www.hydrosheds.org) with values calculated as basin-scale averages weighted by the smallest unit area in the HydroSHEDS database[85]. Hydrological variability was quantified using RAD, defined as the normalized difference between the maximum and minimum long-term average monthly natural discharge values (1971–2000). NSR was defined as the documented number of native fish species within each basin, reflecting resource availability and commonly used in tests of the biotic acceptance and resistance hypotheses[86,87]. RBA was calculated as the watershed area and was used as a quantitative surrogate for habitat size and

heterogeneity. CSI quantifies comprehensive river connectivity, ranging from 0% to 100% (full connectivity), for each individual river reach and accounts for disruptions to the longitudinal, lateral, vertical, and temporal components of connectivity[88]. We calculated the CSI for each river basin as the average CSI across all river reaches within a basin weighted by reach length. Finally, GDP data (at a resolution of 1 square degree) were taken from Nordhaus and Chen[89] and used as a proxy for socioeconomic activity, reflecting introduction pressure associated with transportation and trade fluxes (e.g., pet trade, freight, aquaculture)[90].

Statistical analysis

We applied two modeling approaches tailored to the type of response variable. For the average life-history strategy composition at the basin level (native and non-native fishes separately), we used Hierarchical Modelling of Species Communities (HMSC), because the proportions of the three strategies (equilibrium, opportunistic and periodic) are compositional and therefore interdependent. HMSC is a multivariate Bayesian generalized linear mixed modelling framework that allowed us to account for both correlation among the three strategies included in the models, and spatial autocorrelation[91,92]. For the similarity in life-history strategies between native and non-native fishes, we used generalized linear mixed models (GLMMs).

Since the three life-history strategies proportions sum to one within each basin, we first applied a centered log-ratio (CLR) transformation after zero replacement using the count zero multiplicative method (R package '*compositions*' and '*zCompositions*')[93,94]. Then, the CLR-transformed values were used as multivariate response variables in the HMSC. Explanatory variables included MAT, TS, DIS, RAD, CSI, GDP, NSR, and RBA. The model included random effects for (i) basin spatial coordinates to capture spatial autocorrelation, and (ii) biogeographical realms to account for non-independence among basins within the same realm and for unmeasured biogeographical variation. We did not include interaction terms in the HMSC because we lacked strong a priori hypotheses at the global scale about specific pairwise interactions among predictors and the HMSC already has a complex hierarchical structure, for which a parsimonious specification provides more stable estimates and clearer ecological interpretation. We fitted the models using the R package '*Hmsc*' assuming the default prior distributions[92,95]. Posterior sampling was performed using four Markov chain Monte Carlo

(MCMC) chains, each collecting 250 samples, yielding a total of 1,000 samples. We used a thinning interval of 10,000 and excluded the first 1,250,000 iterations as burn-in, only sampling the subsequent 2,500,000 iterations per chain. MCMC convergence was evaluated using the potential scale reduction factor (Gelman–Rubin statistic) over the parameters related to the fixed effects (β parameters) and the random effects (Ω parameters)[96].

To identify environmental and anthropogenic drivers of similarity in life-history strategies between basin scale native and non-native fishes, we fitted GLMMs with a Gaussian error distribution and logit link function using the R package '*glmmTMB*'[97]. Fixed predictors included MAT, TS, DIS, RAD, CSI, GDP, NSR, and RBA. Biogeographical realms were included as random intercepts to account for non-independence among basins within the same realm and for unmeasured biogeographical variation. In addition, we incorporated a Matern covariance structure based on standardized geographic coordinates of basin centroids to account for residual spatial autocorrelation. We tested interactions between the temperature regime (MAT and TS) and the discharge regime (DIS and RAD) to detect potential non-additive effects of these four dominant environmental variables and selected the best model according to AIC (Supplementary Table S4). In addition, we also tested interactions between each of MAT and TS with GDP to examine whether this key anthropogenic driver modified the main ecological relationships, revealing that GDP did not significantly alter the primary temperature effects (Supplementary Table S8). The explained variance by the fixed effect (marginal R^2) and both the fixed and random effects (conditional R^2) were computed using the R package '*MuMIn*'[98]. We also performed Wilcoxon rank sum tests using the R package '*stats*' to compare the life-history strategy compositions between native and non-native fishes globally and in six biogeographical realms[99]. In addition, we used Kruskal–Wallis tests followed by Dunn's post hoc comparisons (R package '*rstatix*') to assess differences among the three life-history strategies within native and within non-native fishes at both the global and realm scales[100].

All fixed predictors were standardized (z-transformed: subtracting the mean and dividing by the standard deviation) to facilitate comparability of parameter estimates. In the HMSC models, the eight environmental and human-activity variables were overall independent of each other in both datasets (|Spearman's $\rho| < 0.6$), except for MAT and TS in the native dataset ($\rho = -0.73$; Supplementary Fig. S12). Multicollinearity issues in the GLMM were evaluated using

variance inflation factors (VIFs), which were all below four. All statistical analyses were performed using R version 4.4.2 for Windows[101].

Limitations and uncertainties

Although we revealed latitudinal patterns and drivers of global freshwater fish life-history strategies and provided new insights into Darwin's naturalization conundrum based on life-history theory, we acknowledge that our study has several limitations due to data constraints. First, Darwin's naturalization conundrum is rooted in local-scale processes, whereas our analysis focused on global-scale outcomes by examining patterns of similarity in life-history strategies between native and non-native fishes. Future studies should therefore incorporate high-resolution abundance data to provide more robust evidence at regional/local scales. Second, human-mediated introduction biases may also influence global patterns of life-history strategies and similarities between native and non-native species. Although we included GDP as a covariate to approximate introduction pressure, this does not fully account for introduction preferences, such as cross-country or regional differences in the number and identity of species introduced. Third, although using mean values allowed us to compare differences in life-history strategies among basins, it is important to acknowledge that averaging can simplify the variation of life-history strategies within each basin. Finally, due to limited data, we were unable to account for intraspecific variation in life-history traits across different river basins.

Data Availability

The raw data used in this study is publicly available (see Methods). The processed data generated in this study are either included in the Supplementary Information or available in the figshare repository (https://figshare.com/s/a6ce374665806be9d5ca).

Code Availability

The R codes used to conduct the study are available in the GitHub.

Acknowledgements

K.Z. was supported by the National Natural Science Foundation of China (grant No. 32401388), the Postdoctoral Fellowship Program of CPSF (grant No. GZC20241874), the Postdoctoral Project of Hubei Province (grant No. 2024HBBHCXA094) and the China Postdoctoral Science

Foundation (grant No. 2024M753437). J.G.M. was supported by JSPS KAKENHI Grant-in-Aid for Scientific Research (grant No. 25K15516). G.S. was supported by the National Natural Science Foundation of China (grant No. 32470551) and the National Key R&D Program of China (grant No. 2024YFF1307500).

Author Contributions Statement

K.Z. conceived the idea. K.Z., J.X., J.D.O., J.G.M. and O.O. contributed to the study design. K.Z., J.D.O., O.O. and J.G.M. contributed to the acquisition and analysis of data. K.Z., J.D.O., J.G.M., O.O. and G.S. contributed to the interpretation of results. K.Z., O.O., J.D.O., J.G.M., C.L., M.Z., G.S. and J.X. wrote and edited the manuscript.

Competing Interests Statement

All authors have no competing interests.

Correspondence and requests for materials should be addressed to K.Z. or G.S.

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